

Exponential Functions and Their Graphs

ANSWER KEY



Evaluating and describing

- 1 Let $f(x) = 3 \cdot 2^x$. Evaluate:
- a** $f(0) = 3$ **b** $f(3) = 24$ **c** $f(-2) = \frac{3}{4}$
- 2 For $y = 5^x$, state:
- a** the domain All real numbers. **b** the range $y > 0$.
- c** the horizontal asymptote $y = 0$. **d** the y -intercept $(0, 1)$.
- 3 Describe how the graph of $g(x) = 2^{x-3} + 1$ is obtained from the graph of $y = 2^x$, and state the asymptote of g .
- Shift right 3 units and up 1 unit. Horizontal asymptote: $y = 1$.
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Identifying from graphs and applications

- 4 The graph below passes through $(0, 1)$, $(1, 2)$, and $(2, 4)$. Write the equation of the exponential function shown.
- $y = 2^x$.
- 5 An investment of 1000 riyals grows with continuous compounding at an annual rate of 6%:
- $A(t) = 1000e^{0.06t}$. Find the value after 5 years, to the nearest rial.
- $A(5) = 1000e^{0.3} \approx 1350$ riyals.
- 6 Determine whether each function represents growth or decay:
- a** $y = 4(1.08)^x$ Growth. **b** $y = 7(0.85)^x$ Decay.
- c** $y = 2e^{-0.3x}$ Decay. **d** $y = \left(\frac{5}{4}\right)^x$ Growth.
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Compound interest

- 7 2000 riyals is invested at 5% annual interest, compounded monthly, using $A = P\left(1 + \frac{r}{n}\right)^{nt}$.
- a** Find the value after 3 years, to the nearest rial. $A = 2000\left(1 + \frac{0.05}{12}\right)^{36} \approx 2000(1.1616) \approx 2323$ riyals.
- b** Find the value after 3 years if compounded continuously instead ($A = Pe^{rt}$), to the nearest rial.
- $A = 2000e^{0.15} \approx 2000(1.1618) \approx 2324$ riyals.

- 8 How much must be invested now at 4% annual interest compounded annually to have 10,000 riyals in 8 years? Round to the nearest riyal.

$$P = \frac{10000}{(1.04)^8} \approx \frac{10000}{1.3686} \approx 7308 \text{ riyals.}$$

Comparing exponential growth rates

- 9 Two investments start at 500 riyals: one grows at 4% per year, the other at 7% per year. Find the value of each after 10 years, to the nearest riyal, and their difference.

4%: $500(1.04)^{10} \approx 740$ riyals. 7%: $500(1.07)^{10} \approx 984$ riyals. Difference ≈ 244 riyals.

- 10 Rank the following functions from slowest to fastest growth as $x \rightarrow \infty$: $y = x^{10}$, $y = 2^x$, $y = 1.01^x$.

Any exponential with base > 1 eventually outgrows any polynomial, no matter how slowly it starts. So even though 1.01^x grows very slowly at first, it eventually overtakes x^{10} . Long-run ranking (slowest to fastest): $x^{10} < 1.01^x < 2^x$.

Comparing growth and decay graphically

- 11 The graphs of $y = 2^x$ (growth) and $y = (\frac{1}{2})^x$ (decay) are shown together. Use the graph to describe the symmetry relating the two curves.

The two curves are mirror images of each other across the y -axis: $(\frac{1}{2})^x = 2^{-x}$, so reflecting $y = 2^x$ over the y -axis (replacing x with $-x$) produces the decay curve.

The natural exponential base e

- 12 Explain what distinguishes $e \approx 2.71828$ from other exponential bases, in terms of the slope of $y = e^x$ at $x = 0$.

The number e is the unique base for which the exponential function $y = b^x$ has a tangent line slope of EXACTLY 1 at $x = 0$ — i.e., the function's instantaneous rate of change equals its own value at every point, a property no other base shares.

- 13 Evaluate e^0 , e^1 (to 3 decimal places), and simplify $e^{\ln 5}$.

$e^0 = 1$. $e^1 \approx 2.718$. $e^{\ln 5} = 5$ (since e^x and $\ln x$ are inverse functions).

Piecewise-exponential applications

- 14 A machine depreciates in value according to $V(t) = 15000(0.88)^t$ (riyals, years). Find its value after 5 years, and the year in which its value first drops below 5000 riyals (to the nearest year).

$V(5) = 15000(0.88)^5 \approx 15000(0.5277) \approx 7916$ riyals. Setting $15000(0.88)^t < 5000$:

$(0.88)^t < 0.333$. Testing values: at $t = 9$, $(0.88)^9 \approx 0.316 < 0.333$; at $t = 8$, $(0.88)^8 \approx 0.360 > 0.333$. First drops below 5000 riyals in year 9.

Transformations of exponential functions

15 State the horizontal asymptote and range of each transformed exponential function:

a $y = 3^x + 4$ Asymptote $y = 4$; range $y > 4$.

b $y = -2^x$ Asymptote $y = 0$; range $y < 0$ (reflected below the axis).

c $y = 5^x - 3$ Asymptote $y = -3$; range $y > -3$.

16 Write the equation of the function obtained by taking $y = e^x$, reflecting it across the y -axis, then shifting it down 2 units.

Reflecting across the y -axis replaces x with $-x$: $y = e^{-x}$. Shifting down 2: $y = e^{-x} - 2$.

Comparing exponential functions numerically

17 For $f(x) = 3^x$, complete a short table of values at $x = -1, 0, 1, 2, 3$ and verify each value satisfies $f(x+1) = 3f(x)$.

$f(-1) = \frac{1}{3}$, $f(0) = 1$, $f(1) = 3$, $f(2) = 9$, $f(3) = 27$. Check: $f(0) = 3f(-1) = 3(1/3) = 1$ ✓;

$f(1) = 3f(0) = 3$ ✓; $f(2) = 3f(1) = 9$ ✓; $f(3) = 3f(2) = 27$ ✓ — each step multiplies by the base 3, confirming the defining property of exponential growth.